



Name _____ HR _____

3

8th Math Unit 3 - Linear Algebra STAR Cards!

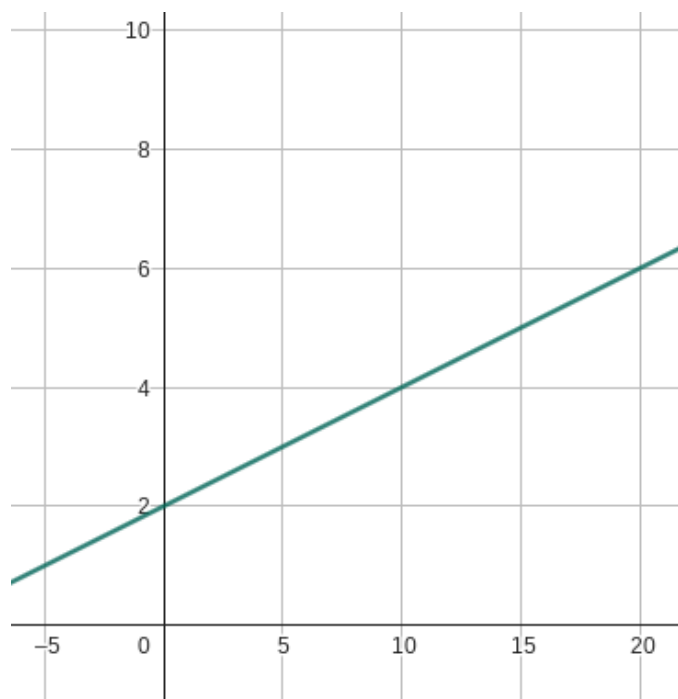
QUESTION	ANSWER
3a) ☆☆☆ What are the 3 main ways to represent an algebraic relationship?	3a) <u>G</u>raph, <u>E</u>quation, <u>T</u>able
3b) ☆☆☆ 1) What is the “recipe” for an equation that represents a linear relationship? 2) What letter is the slope? 3) What letter is the starting point?	3b) 1) $y = mx + b$ <i>Starting Point</i> 2) “M” is the slope 3) “B” is the starting point

QUESTION

3c)



1. Explain how you can find the slope from the graph below.
2. Explain how you find the starting point from the graph below.



ANSWER

3c)

1.
 - a. Find two points on the graph
 - b. Count rise (how far up or down)
 - c. Count run (how far left or right)
 - d. Write slope as fraction: $\frac{\Delta y}{\Delta x}$

2. Find the number where the graphed line crosses **y-axis**

Example: in this picture (0,2)

3d)

1. Explain how you find the slope from the table below?
2. Explain how you find the starting point from the table below

Example)

x	y
0	4
3	6
6	8
8	10

3d)

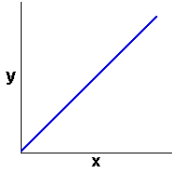
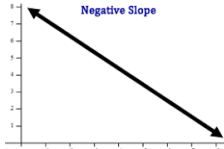
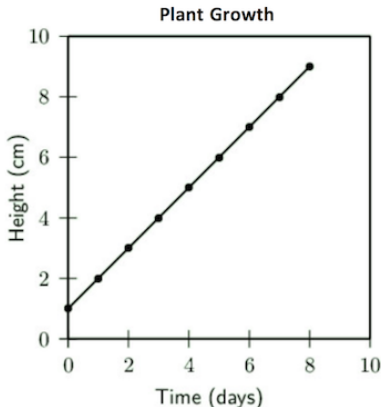
1. Make a fraction showing example:

The change in y values
The change in x values

x	y
0	4
3	6
6	8
8	10

$$\text{Slope} = \frac{2}{3}$$

2. The number next to when $x = 0$
Starting point: 4

QUESTION	ANSWER
3e) ☆☆☆ 1) How can you tell if a line on a graph shows a positive relationship? 2) How can you tell if a line on a graph shows a negative relationship?	3e) 1) Positive: going up to the right  2) Negative: going down to the right 
3f) ☆☆☆ In the situation graphed below: 1) What does the slope represent? 2) What does the starting point represent? 	3f) 1) The slope shows how much the plant is growing over time: $\frac{\text{height (cm)}}{\text{time(days)}}$ 2) The height of the plant at day zero
3g) ☆☆☆ What does the graph of this equation look like? $y = \frac{1}{3}x + 2$	3g) A straight line that: • touches the y-axis at 2 • has a slope of $\frac{1}{3}$ (every time it goes up by 1, it goes over by 3) 